

DYNAMICAL SYSTEMS

PROBLEM SET 1

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We begin by noting that all of the files referenced in this document are available at:
<https://github.com/chtompki/MathFall2026/tree/main/DynamicalSystems/ProblemSet1>

Part 2: from Strogatz, 2nd Edition, Chapter 2

2.2.4 Analyze the following equation graphically. Sketch the vector field on the real line find all the fixed points, classify their stability, and sketch the graph of $x(t)$ for different initial conditions. Then try to find the analytical solution for $x(t)$, or state that it's impossible. The equation in question is:

$$\dot{x} = e^{-x} \sin(x)$$

Solution. Set

$$f(x) = e^{-x} \sin x.$$

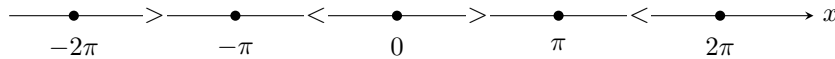
Because $e^{-x} > 0$ for every $x \in \mathbb{R}$, the sign of $f(x)$ is the same as the sign of $\sin x$. The fixed points are therefore

$$f(x_*) = 0 \iff \sin x_* = 0 \iff \boxed{x_* = n\pi, \quad n \in \mathbb{Z}.}$$

Moreover,

$$\dot{x} > 0 \quad \text{on } (2k\pi, (2k+1)\pi), \quad \dot{x} < 0 \quad \text{on } ((2k-1)\pi, 2k\pi).$$

Thus a representative portion of the phase line is



For the stability classification, differentiate:

$$f'(x) = e^{-x}(\cos x - \sin x).$$

At a fixed point $x_* = n\pi$,

$$f'(n\pi) = (-1)^n e^{-n\pi}.$$

Consequently,

$$\boxed{x = 2k\pi \text{ is unstable,} \quad x = (2k+1)\pi \text{ is asymptotically stable.}}$$

All the fixed points are hyperbolic.

Now let $x(0) = x_0$. If $x_0 = n\pi$, uniqueness gives the equilibrium solution

$$x(t) \equiv n\pi.$$

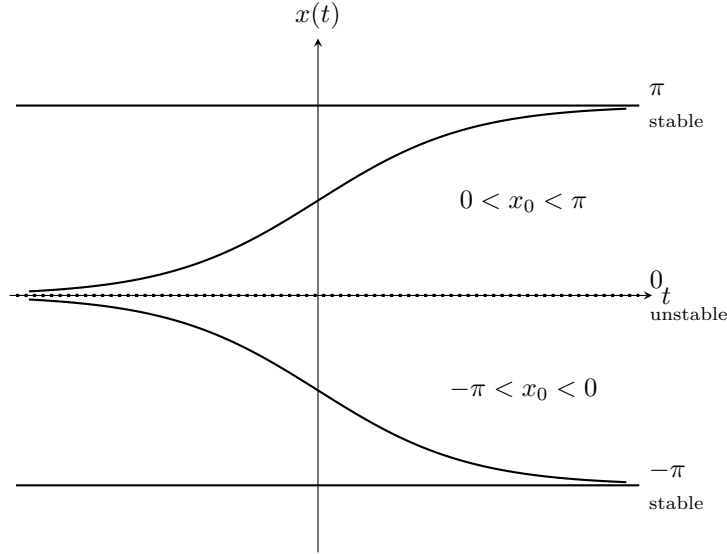
If $2k\pi < x_0 < (2k+1)\pi$, then $x(t)$ increases monotonically and

$$\lim_{t \rightarrow -\infty} x(t) = 2k\pi, \quad \lim_{t \rightarrow \infty} x(t) = (2k+1)\pi.$$

If $(2k-1)\pi < x_0 < 2k\pi$, then $x(t)$ decreases monotonically and

$$\lim_{t \rightarrow -\infty} x(t) = 2k\pi, \quad \lim_{t \rightarrow \infty} x(t) = (2k-1)\pi.$$

Thus every nonconstant orbit connects an unstable even multiple of π to one of the adjacent stable odd multiples. Representative solution curves near $x = 0$ are shown below.



The drawn curves are qualitative sketches; they indicate the correct monotonicity and limiting equilibria, not an explicit formula for the solutions.

For a nonconstant solution, separation of variables yields

$$\frac{e^x}{\sin x} dx = dt.$$

Hence, as long as $x(t)$ remains in the interval between the neighboring fixed points containing x_0 ,

$$t = \int_{x_0}^{x(t)} \frac{e^u}{\sin u} du.$$

The antiderivative $\int e^x \csc x dx$ is not an elementary function, so there is no elementary closed-form expression for $x(t)$. If

$$F(x) = \int \frac{e^u}{\sin u} du$$

is defined on the interval containing x_0 , then the solution may be written implicitly, or in inverse-function form, as

$$F(x(t)) - F(x_0) = t, \quad \boxed{x(t) = F^{-1}(t + F(x_0))}.$$

Together with the constant solutions $x(t) \equiv n\pi$, this describes all solutions. □

2.2.7 Analyze the following equation graphically. Sketch the vector field on the real line find all the fixed points, classify their stability, and sketch the graph of $x(t)$ for different initial conditions. Then try to find the analytical solution for $x(t)$, or state that it's impossible. The equation in question is:

$$\dot{x} = e^x - \cos(x)$$

(Hint: Sketch the graphs of e^x and $\cos(x)$ on the same axes, and look for intersections. You won't be able to find the fixed points explicitly, but you still can find the qualitative behaviour.)

Solution. Set

$$f(x) = e^x - \cos x.$$

Fixed points are intersections of $y = e^x$ and $y = \cos x$. For $x > 0$, $e^x > 1 \geq \cos x$, while $e^0 = \cos 0 = 1$. Hence $x = 0$ is the only nonnegative fixed point.

For $x < 0$, an intersection is possible only where $\cos x > 0$. On every positive cosine lobe centered at $-2n\pi$, $n \geq 1$, one has

$$f\left(-\left(2n + \frac{1}{2}\right)\pi\right) > 0, \quad f(-2n\pi) = e^{-2n\pi} - 1 < 0,$$

and

$$f\left(-\left(2n - \frac{1}{2}\right)\pi\right) > 0.$$

Therefore there are two fixed points on each such lobe. There is also one fixed point in $(-\pi/2, 0)$, in addition to the fixed point at 0.

Write the negative fixed points as

$$a_0 \in \left(-\frac{\pi}{2}, 0\right)$$

and, for $n \geq 1$,

$$a_n \in \left(-\left(2n + \frac{1}{2}\right)\pi, -2n\pi\right), \quad b_n \in \left(-2n\pi, -\left(2n - \frac{1}{2}\right)\pi\right).$$

Thus all the fixed points are

$$\boxed{0, \quad a_0, \quad a_n, \quad b_n \quad (n \geq 1).}$$

They cannot be expressed in elementary closed form. Numerically, the first few are

$$a_0 \approx -1.29270, \quad b_1 \approx -4.72129, \quad a_1 \approx -7.85359, \quad b_2 \approx -10.99559.$$

As one moves from left to right across a positive cosine lobe, the sign of f changes from positive to negative at a_n , and from negative to positive at b_n . The same positive-to-negative change occurs at a_0 , whereas f changes from negative to positive at 0. A representative portion of the phase line is therefore

$$\begin{array}{ccccccc} \text{U} & & \text{S} & & \text{U} & & \text{S} & & \text{U} \\ \bullet & \xrightarrow{\quad} & \bullet & \xleftarrow{\quad} & \bullet & \xrightarrow{\quad} & \bullet & \xleftarrow{\quad} & \bullet \\ b_2 & & a_1 & & b_1 & & a_0 & & 0 \end{array} \rightarrow x$$

Here S denotes asymptotically stable and U denotes unstable. Consequently,

$$\boxed{a_n \text{ is asymptotically stable for every } n \geq 0,}$$

whereas

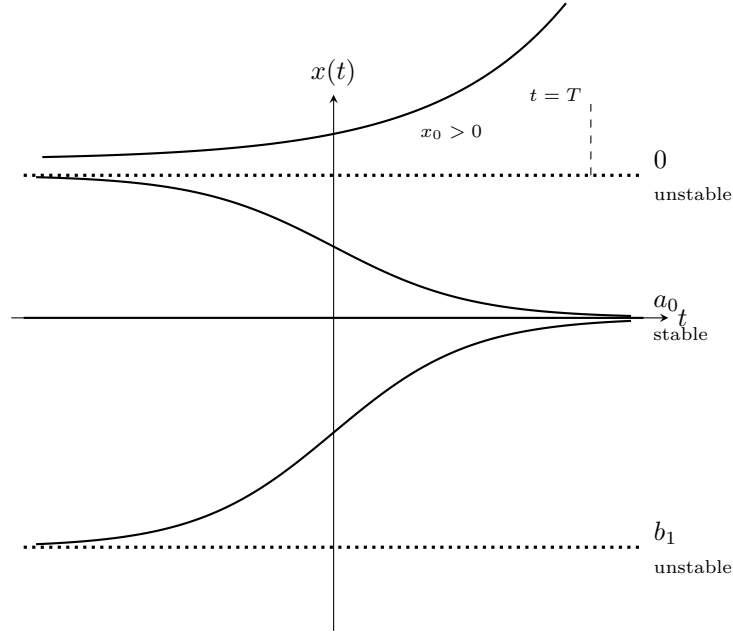
$$\boxed{b_n \text{ is unstable for every } n \geq 1, \quad \text{and } x = 0 \text{ is unstable.}}$$

In particular, $f'(0) = e^0 + \sin 0 = 1 > 0$, confirming directly that 0 is unstable.

The phase line determines all the solution behavior. Between any unstable fixed point and the stable fixed point immediately to its right, solutions increase toward that stable point. Between a stable fixed point and the unstable fixed point immediately to its right, solutions decrease toward the stable point. For example,

$$\begin{aligned} b_1 < x_0 < a_0 : \quad x(t) \nearrow a_0 \quad (t \rightarrow \infty), \quad x(t) \rightarrow b_1 \quad (t \rightarrow -\infty), \\ a_0 < x_0 < 0 : \quad x(t) \searrow a_0 \quad (t \rightarrow \infty), \quad x(t) \rightarrow 0 \quad (t \rightarrow -\infty). \end{aligned}$$

Initial data at a fixed point give a constant solution. Representative solution curves are shown below; solid horizontal lines are stable equilibria and dotted horizontal lines are unstable equilibria.



The curves are qualitative sketches rather than exact solution formulas. For $x_0 > 0$, $f(x) > 0$, so the solution increases without bound. In fact it reaches $+\infty$ in finite forward time because, for sufficiently large x ,

$$e^x - \cos x \geq \frac{1}{2}e^x, \quad \int^{\infty} \frac{dx}{e^x - \cos x} < \infty.$$

Thus $x(t) \rightarrow +\infty$ as $t \uparrow T$ for some finite T .

Finally, every nonconstant solution is separable:

$$\frac{dx}{e^x - \cos x} = dt.$$

With $x(0) = x_0$, its exact implicit form is

$$t = \int_{x_0}^{x(t)} \frac{du}{e^u - \cos u}.$$

The antiderivative is not elementary, so no elementary explicit formula for $x(t)$ is available. On any interval that contains no fixed point, define

$$F(x) = \int \frac{du}{e^u - \cos u}.$$

Then the solution can equivalently be written in inverse-function form as

$$x(t) = F^{-1}(t + F(x_0)).$$

Together with the constant solutions at all the fixed points, this describes all solutions. □

2.2.8 (Working backwards, from flows to equations) Given an equation

$$\dot{x} = f(x),$$

we know how to sketch the corresponding flow on the real line. Here you are asked to solve the opposite problem: For the phase portrait shown in Figure 1, find an equation that is consistent with it. (There are an infinite number of correct answers—and wrong ones too.)

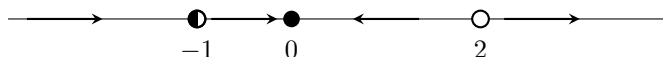


FIGURE 1.

Solution. The fixed points in the portrait are

$$x = -1, \quad x = 0, \quad \text{and} \quad x = 2.$$

The arrows prescribe the following sign pattern for f :

x	$(-\infty, -1)$	$(-1, 0)$	$(0, 2)$	$(2, \infty)$
$f(x)$	+	+	-	+

At $x = -1$, the sign does not change. Thus the simplest polynomial should have a zero of even multiplicity there; we choose the factor $(x + 1)^2$. At both $x = 0$ and $x = 2$, the sign changes, so we choose simple factors x and $(x - 2)$. This suggests

$$f(x) = (x + 1)^2 x(x - 2).$$

Indeed, $(x + 1)^2 \geq 0$, so away from $x = -1$ the sign of f is determined by $x(x - 2)$. Therefore

$$x < 0: \quad x(x - 2) > 0,$$

$$0 < x < 2: \quad x(x - 2) < 0,$$

$$x > 2: \quad x(x - 2) > 0.$$

This gives precisely the required arrows on every interval, including the positive sign on both sides of $x = -1$. Hence one equation consistent with the portrait is

$$\dot{x} = (x + 1)^2 x(x - 2).$$

The stability shown by this equation also agrees with the portrait:

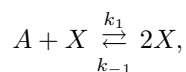
- At $x = -1$, solutions move toward the fixed point from the left and away from it on the right. Thus $x = -1$ is half-stable (stable from the left and unstable from the right).
- At $x = 0$, arrows point toward the fixed point from both sides, so $x = 0$ is asymptotically stable.
- At $x = 2$, arrows point away from the fixed point on both sides, so $x = 2$ is unstable.

The answer is not unique. More generally, for any continuous function $g(x) > 0$ on $\mathbb{R}$,

$$\dot{x} = g(x)(x + 1)^2 x(x - 2)$$

has the same fixed points and the same phase portrait. One could also use higher odd multiplicities at 0 or 2, or a higher even multiplicity at -1 , provided no additional zeros are introduced and the required sign pattern is preserved. $\square$

2.3.2 (Autocatalysis) Consider the model chemical reaction



in which one molecule of X combines with one molecule of A to form two molecules of X . This means that the chemical X stimulates its own production, a process called *autocatalysis*. This positive feedback process leads to a chain reaction, which eventually is limited by a “back reaction” in which $2X$ returns to $A + X$.

According to the *law of mass action* of chemical kinetics, the rate of an elementary reaction is proportional to the product of the concentrations of the reactants. We denote the concentrations by lowercase letters $x = [X]$ and $a = [A]$. Assume that there is an enormous surplus of chemical A , so that its concentration a can be regarded as constant. Then the equation for the kinetics of x is

$$\dot{x} = k_1 ax - k_{-1} x^2,$$

where k_1 and k_{-1} are positive parameters called *rate constants*.

- a) Find all the fixed points of this equation and classify their stability.
- b) Sketch the graph of $x(t)$ for various initial values x_0 .

Solution. It is convenient to introduce

$$r = k_1 a > 0, \quad K = \frac{k_1 a}{k_{-1}} > 0.$$

Then the equation becomes the logistic equation

$$\dot{x} = rx \left(1 - \frac{x}{K}\right).$$

- a) The fixed points satisfy

$$rx \left(1 - \frac{x}{K}\right) = 0.$$

Therefore

$$\boxed{x_1^* = 0, \quad x_2^* = K = \frac{k_1 a}{k_{-1}}.}$$

Let

$$f(x) = k_1 ax - k_{-1} x^2.$$

Its derivative is

$$f'(x) = k_1 a - 2k_{-1} x.$$

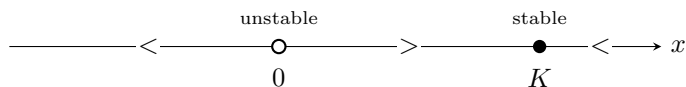
At the fixed points,

$$f'(0) = k_1 a > 0, \quad f'(K) = k_1 a - 2k_{-1} \frac{k_1 a}{k_{-1}} = -k_1 a < 0.$$

Hence

$$\boxed{x = 0 \text{ is unstable,} \quad x = K \text{ is asymptotically stable.}}$$

The corresponding phase line is

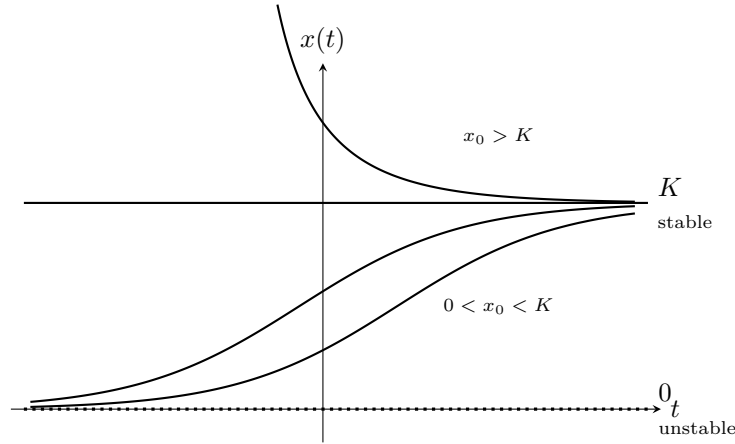


Only $x \geq 0$ is physically meaningful because x is a concentration. On this physical domain, solutions move away from 0 and toward K .

b) Let $x(0) = x_0$. The qualitative behavior is as follows:

Initial value	Behavior as $t \rightarrow \infty$
$x_0 = 0$	$x(t) \equiv 0$,
$0 < x_0 < K$	$x(t)$ increases monotonically to K ,
$x_0 = K$	$x(t) \equiv K$,
$x_0 > K$	$x(t)$ decreases monotonically to K .

Representative solution curves are shown below. The stable equilibrium is solid and the unstable equilibrium is dotted.



For completeness, the equation can also be solved analytically. Separating variables gives

$$\frac{dx}{x(1 - x/K)} = r dt.$$

Since

$$\frac{1}{x(1 - x/K)} = \frac{1}{x} + \frac{1}{K - x},$$

integration yields

$$\ln \left| \frac{x}{K - x} \right| = rt + C.$$

For $x_0 \neq 0$, solving for $x(t)$ gives

$$x(t) = \frac{K}{1 + \left(\frac{K}{x_0} - 1 \right) e^{-rt}} = \frac{\frac{k_1 a}{k_{-1}}}{1 + \left(\frac{k_1 a}{k_{-1} x_0} - 1 \right) e^{-k_1 a t}}.$$

The case $x_0 = 0$ must be included separately and gives $x(t) \equiv 0$.

Mathematically, if $x_0 < 0$, then $\dot{x} < 0$ and the solution decreases toward $-\infty$ in finite forward time. Such solutions are not chemically meaningful, since a concentration cannot be negative.

□

2.3.3 (Tumor growth) The growth of cancerous tumors can be modeled by the Gompertz law

$$\dot{N} = -aN \ln(bN),$$

where $N(t)$ is proportional to the number of cells in the tumor, and $a, b > 0$ are parameters.

- a) Interpret a and b biologically.
- b) Sketch the vector field and graph $N(t)$ for various initial values.

The predictions of this simple model agree surprisingly well with data on tumor growth, as long as N is not too small; see Aroesty et al. (1973) and Newton (1980) for examples.

Solution. Because N represents tumor size, the physically meaningful domain is $N \geq 0$. The displayed differential equation is defined for $N > 0$; since

$$\lim_{N \rightarrow 0^+} N \ln(bN) = 0,$$

we may continuously extend its right-hand side by defining $\dot{N} = 0$ at $N = 0$.

- a) Introduce the limiting tumor size

$$K = \frac{1}{b}.$$

Then the Gompertz law can be written as

$$\dot{N} = aN \ln\left(\frac{K}{N}\right).$$

The parameter b therefore determines the saturation size, or carrying capacity,

$$N_\infty = K = \frac{1}{b}.$$

Thus increasing b decreases the predicted limiting tumor size.

The per-capita growth rate is

$$\frac{\dot{N}}{N} = a \ln\left(\frac{K}{N}\right).$$

Hence a sets the time scale on which the tumor approaches its limiting size. In fact, as shown below, the logarithmic distance from the limiting size decays like e^{-at} , so the characteristic time is

$$\tau = \frac{1}{a}.$$

Larger a means faster growth and faster relaxation toward K .

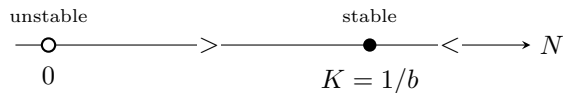
- b) For $N > 0$, let

$$f(N) = -aN \ln(bN) = aN \ln\left(\frac{K}{N}\right).$$

The fixed points on the extended physical domain are

$$N_1^* = 0, \quad N_2^* = K = \frac{1}{b}.$$

For $0 < N < K$, one has $\ln(K/N) > 0$, so $\dot{N} > 0$. For $N > K$, $\ln(K/N) < 0$, so $\dot{N} < 0$. The phase line is therefore

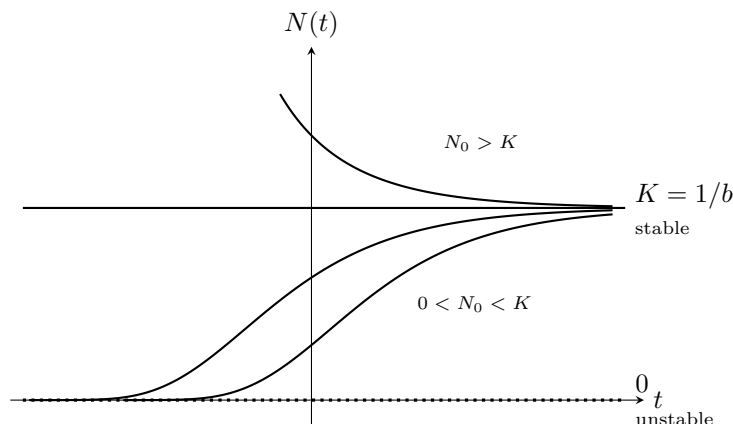


Thus $N = 0$ is unstable from the physically relevant side, while $N = K$ is asymptotically stable. In particular,

$$\begin{array}{ll} 0 < N_0 < K & : N(t) \text{ increases toward } K, \\ N_0 = K & : N(t) \equiv K, \\ N_0 > K & : N(t) \text{ decreases toward } K. \end{array}$$

Also, under the continuous extension at the origin, $N_0 = 0$ gives $N(t) \equiv 0$.

Representative solution curves are shown below. The unstable equilibrium is dotted and the stable equilibrium is solid.



The solution can be found explicitly. Set

$$y(t) = \ln(bN(t)).$$

For $N(t) > 0$,

$$\dot{y} = \frac{\dot{N}}{N} = -a \ln(bN) = -ay.$$

Therefore

$$y(t) = y(0)e^{-at} = \ln(bN_0)e^{-at}.$$

Exponentiating gives

$$N(t) = \frac{1}{b} \exp(\ln(bN_0)e^{-at}), \quad N_0 > 0.$$

Equivalently,

$$N(t) = K \exp\left(\ln\left(\frac{N_0}{K}\right)e^{-at}\right).$$

This formula shows directly that $N(t) \rightarrow K = 1/b$ as $t \rightarrow \infty$ for every $N_0 > 0$.

For a growing solution, the curvature changes sign when $N = K/e$. Thus a solution starting below K/e is initially concave upward and later concave downward, giving the characteristic sigmoidal Gompertz growth curve. If $K/e \leq N_0 < K$, it is concave downward for all $t \geq 0$.

□

2.3.5 (Dominance of the fittest) Suppose X and Y are two species that reproduce exponentially fast: $\dot{X} = aX$ and $\dot{Y} = bY$, respectively, with initial conditions $X_0, Y_0 > 0$ and growth rates $a \geq b \geq 0$. Here X is “fitter” than Y in the sense that it reproduces faster, as reflected by the inequality $a > b$. So we’d expect X to keep increasing its share of the total population $X + Y$ as $t \rightarrow \infty$. The goal of this exercise is to demonstrate this intuitive result, first analytically and then geometrically.

- Let $x(t) = X(t)/[X(t) + Y(t)]$ denote X ’s share of the total population. By solving for $X(t)$ and $Y(t)$, show that $x(t)$ increasing monotonically approaches 1 as $t \rightarrow \infty$.
- Alternatively, we can arrive at the same conclusions by deriving a differential equation for $x(t)$. To do so, take the time derivative of $x(t) = X(t)/[X(t) + Y(t)]$ using the quotient and chain rules. Then

substitute for X and Y and thereby show that $x(t)$ obeys the logistic equation $\dot{x} = (a - b)x(1 - x)$. Explain why this implies that $x(t)$ increases monotonically and approaches 1 as $t \rightarrow \infty$.

Solution. (a) The equations for the two populations are independent exponential-growth equations. Their solutions are

$$X(t) = X_0 e^{at}, \quad Y(t) = Y_0 e^{bt}.$$

Consequently,

$$\begin{aligned} x(t) &= \frac{X_0 e^{at}}{X_0 e^{at} + Y_0 e^{bt}} \\ &= \frac{1}{1 + \frac{Y_0}{X_0} e^{-(a-b)t}}. \end{aligned}$$

Since $X_0, Y_0 > 0$, we have

$$0 < x(t) < 1$$

for every finite t . Moreover, because $a > b$, the function $e^{-(a-b)t}$ decreases strictly as t increases. Therefore, the denominator

$$1 + \frac{Y_0}{X_0} e^{-(a-b)t}$$

decreases strictly, so $x(t)$ increases strictly.

Finally, $a - b > 0$, and hence

$$\lim_{t \rightarrow \infty} e^{-(a-b)t} = 0.$$

It follows that

$$\boxed{\lim_{t \rightarrow \infty} x(t) = 1.}$$

Thus, the fitter species X monotonically increases its share of the total population, and its share approaches 100%.

(b) Applying the quotient rule gives

$$\begin{aligned} \dot{x} &= \frac{\dot{X}(X + Y) - X(\dot{X} + \dot{Y})}{(X + Y)^2} \\ &= \frac{\dot{X}Y - X\dot{Y}}{(X + Y)^2}. \end{aligned}$$

Substituting $\dot{X} = aX$ and $\dot{Y} = bY$, we obtain

$$\dot{x} = \frac{aXY - bXY}{(X + Y)^2} = (a - b) \frac{XY}{(X + Y)^2}.$$

From the definition of x ,

$$x = \frac{X}{X + Y} \quad \text{and} \quad 1 - x = \frac{Y}{X + Y}.$$

Therefore,

$$x(1 - x) = \frac{X}{X + Y} \frac{Y}{X + Y} = \frac{XY}{(X + Y)^2}.$$

Substitution yields the logistic equation

$$\boxed{\dot{x} = (a - b)x(1 - x).}$$

Because $X, Y > 0$, the population share satisfies $0 < x < 1$. Since $a - b > 0$, it follows that

$$\dot{x} = (a - b)x(1 - x) > 0 \quad \text{whenever } 0 < x < 1.$$

Thus, $x(t)$ increases strictly and is bounded above by 1, so it must approach a limit $L \leq 1$. The equilibria of the logistic equation are determined by

$$(a - b)L(1 - L) = 0,$$

giving $L = 0$ or $L = 1$. Since $x(t) > 0$ and is increasing, it cannot approach 0. Hence,

$$\boxed{\lim_{t \rightarrow \infty} x(t) = 1.}$$

Geometrically, the phase line for $0 < x < 1$ points to the right because $\dot{x} > 0$. Therefore, every solution beginning in this interval moves monotonically toward the stable equilibrium $x = 1$. □

2.4.3 Use linear stability analysis to classify the fixed points of the following system. If linear stability analysis fails because $f'(x^*) = 0$, use a graphical argument to decide the stability.

$$\dot{x} = \tan(x)$$

Solution. Let

$$f(x) = \tan(x).$$

The fixed points x^* satisfy

$$f(x^*) = 0.$$

Since

$$\tan(x) = 0 \iff x = n\pi, \quad n \in \mathbb{Z},$$

the system has the fixed points

$$\boxed{x_n^* = n\pi, \quad n \in \mathbb{Z}.}$$

To classify them, we compute

$$f'(x) = \sec^2(x).$$

At each fixed point,

$$f'(x_n^*) = \sec^2(n\pi) = 1 > 0.$$

For a one-dimensional autonomous system, a fixed point is unstable when $f'(x^*) > 0$. Therefore, every fixed point $x_n^* = n\pi$ is unstable:

$$\boxed{\text{Every fixed point } x_n^* = n\pi \text{ is unstable.}}$$

To see this directly, consider a small perturbation

$$x = n\pi + \eta.$$

The linearized equation is

$$\dot{\eta} = f'(n\pi)\eta = \eta,$$

whose solution is

$$\eta(t) = \eta(0)e^t.$$

Thus, any nonzero perturbation grows exponentially, causing the solution to move away from $n\pi$.

This classification is also evident from the phase line. Immediately to the left of $n\pi$,

$$\tan(x) < 0,$$

so solutions move to the left. Immediately to the right of $n\pi$,

$$\tan(x) > 0,$$

so solutions move to the right. Hence, the arrows point away from each fixed point:

$$\longleftarrow \quad n\pi \quad \longrightarrow$$

confirming that every $n\pi$ is a repelling, unstable fixed point.

The points

$$x = \frac{\pi}{2} + n\pi, \quad n \in \mathbb{Z},$$

are not fixed points; they are singularities at which $\tan(x)$ and the differential equation are undefined. □

2.4.8 using linear stability analysis, classify the fixed points of the Gompertz model of tumor growth $\dot{N} = -aN \ln(bN)$. (Similar to Exercise 2.3.3, $N(t)$ is proportional to the number of cells in the tumor and $a, b > 0$ are parameters.)

Solution. Let

$$f(N) = -aN \ln(bN).$$

For $N > 0$, a fixed point N^* satisfies

$$-aN^* \ln(bN^*) = 0.$$

Since $a > 0$ and $N^* > 0$, this requires

$$\ln(bN^*) = 0.$$

Therefore,

$$bN^* = 1,$$

and the positive fixed point is

$$\boxed{N^* = \frac{1}{b}}.$$

To determine its stability, differentiate f :

$$\begin{aligned} f'(N) &= -a \frac{d}{dN} [N \ln(bN)] \\ &= -a [\ln(bN) + 1]. \end{aligned}$$

At $N^* = 1/b$,

$$f' \left(\frac{1}{b} \right) = -a [\ln(1) + 1] = -a < 0.$$

Thus, linear stability analysis shows that

$$\boxed{N^* = \frac{1}{b} \text{ is asymptotically stable.}}$$

Indeed, if $N = 1/b + \eta$, where η is a small perturbation, then the linearized equation is

$$\dot{\eta} = f' \left(\frac{1}{b} \right) \eta = -a\eta.$$

Its solution is

$$\eta(t) = \eta(0)e^{-at},$$

so perturbations decay exponentially.

There is a minor issue at $N = 0$, since $\ln(bN)$ is not defined there. However,

$$\lim_{N \rightarrow 0^+} N \ln(bN) = 0,$$

so the vector field can be continuously extended by defining $f(0) = 0$. Under this biologically natural extension, $N = 0$ is also a fixed point.

Linear stability analysis cannot be applied at $N = 0$, because

$$f'(N) = -a[\ln(bN) + 1] \longrightarrow +\infty \quad \text{as } N \rightarrow 0^+.$$

Instead, consider the sign of $f(N)$. For

$$0 < N < \frac{1}{b},$$

we have $0 < bN < 1$, so $\ln(bN) < 0$. Consequently,

$$\dot{N} = -aN \ln(bN) > 0.$$

Thus, every sufficiently small positive population moves away from $N = 0$. Hence,

$$\boxed{N^* = 0 \text{ is unstable from the biologically relevant side } N > 0.}$$

The phase line is therefore

$$0 \longrightarrow \frac{1}{b} \longleftarrow,$$

because

$$\dot{N} \begin{cases} > 0, & 0 < N < 1/b, \\ < 0, & N > 1/b. \end{cases}$$

Thus, all positive solutions move toward the stable tumor size $N = 1/b$.

□

2.8.2 Sketch the slope field for the following differential equations. Then “integrate” the equation manually by drawing trajectories that are everywhere parallel to the local slope.

(a) $\dot{x} = x$

(c) $\dot{x} = 1 - 4x(1 - x)$

Solution. In each plot, t is the horizontal coordinate and x is the vertical coordinate. Because both equations are autonomous, the slope depends only on x . Thus all line elements on a given horizontal line have the same slope.

(a) Here the local slope is

$$\frac{dx}{dt} = x.$$

The slope is zero on $x = 0$, positive above $x = 0$, and negative below it. Its magnitude increases as $|x|$ increases. Separating variables (away from $x = 0$) gives

$$\frac{dx}{x} = dt, \quad \ln |x| = t + C,$$

so the trajectories are

$$\boxed{x(t) = Ce^t}, \quad C \in \mathbb{R}.$$

The case $C = 0$ is the equilibrium solution $x(t) = 0$. If $C > 0$, the trajectory rises exponentially as t increases; if $C < 0$, it falls farther below zero. Hence $x^* = 0$ is an unstable equilibrium. These exponential curves are tangent to the corresponding line elements of the slope field at every point.

(c) First simplify the vector field:

$$\frac{dx}{dt} = 1 - 4x(1 - x) = 4x^2 - 4x + 1 = (2x - 1)^2 = 4 \left(x - \frac{1}{2}\right)^2.$$

Therefore the slope is nonnegative everywhere and vanishes only on $x = \frac{1}{2}$. The horizontal line

$$x(t) = \frac{1}{2}$$

is an equilibrium trajectory. All other trajectories increase with t . The slope becomes flatter near $x = \frac{1}{2}$ and steeper farther away.

For a non-equilibrium solution, set $y = x - \frac{1}{2}$. Then

$$\dot{y} = 4y^2, \quad \frac{dy}{y^2} = 4 dt.$$

Integration yields

$$-\frac{1}{y} = 4t + C,$$

or, in terms of an initial condition $x(t_0) = x_0 \neq \frac{1}{2}$,

$$x(t) = \frac{1}{2} + \frac{x_0 - \frac{1}{2}}{1 - 4(x_0 - \frac{1}{2})(t - t_0)}.$$

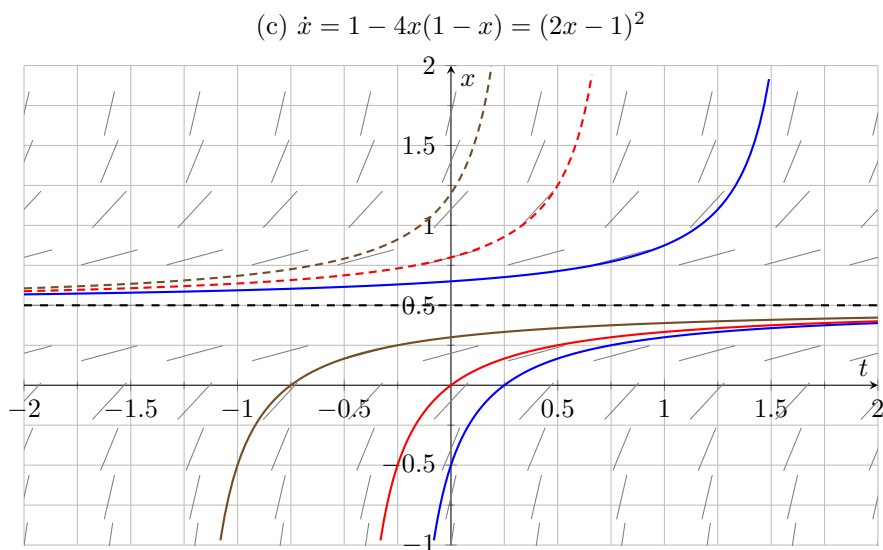
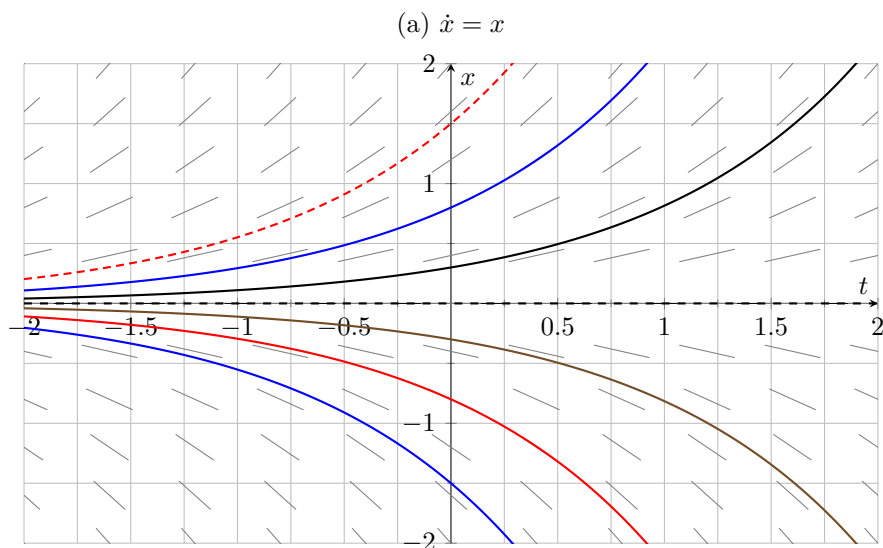
If $x_0 < \frac{1}{2}$, the solution increases toward $\frac{1}{2}$ as $t \rightarrow \infty$. If $x_0 > \frac{1}{2}$, it increases away from the equilibrium and becomes unbounded at the finite time

$$t = t_0 + \frac{1}{4(x_0 - \frac{1}{2})}.$$

Thus $x^* = \frac{1}{2}$ attracts trajectories from below but repels them from above: it is *semistable*. The phase-line arrows point upward on both sides of $x = \frac{1}{2}$.

SLOPE FIELDS AND REPRESENTATIVE TRAJECTORIES

In each panel, the short gray segments form the slope field and the colored curves are representative solution trajectories. Dashed black lines denote equilibrium solutions. The horizontal coordinate is t and the vertical coordinate is x .



In part (c), trajectories below $x = \frac{1}{2}$ approach the equilibrium, whereas trajectories above it move away and become unbounded in finite forward time. This visually confirms that $x = \frac{1}{2}$ is semistable. $\square$

2.8.6 (Analytically intractable problem) Consider the initial value problem $\dot{x} = x + e^{-x}$, $x(0) = 0$.

- Sketch the solution $x(t)$ for $t \geq 0$
- Using some analytical arguments, obtain rigorous bounds on the value of x at $t = 1$. In other words prove that $a < x(1) < b$, for a, b to be determined. By being clever, try to make a and b as close together as possible. (Hint: Bound the given vector field by approximate vector fields that can be integrated analytically.)

- (c) Now for the numerical part: Using the Euler method, compute x at $t = 1$, correct to three decimal places. How small does the stepsize need to be to obtain the desired accuracy? (Give the order of magnitude, not the exact number)

Solution. Write $f(x) = x + e^{-x}$. Since $f(x) > 0$ for every x , the solution is strictly increasing. Also

$$\ddot{x} = f'(x)\dot{x} = (1 - e^{-x})(x + e^{-x}).$$

For $t > 0$ we have $x(t) > 0$, so $1 - e^{-x(t)} > 0$ and consequently $\ddot{x}(t) > 0$. Thus the solution starts at the origin with slope $\dot{x}(0) = 1$, and is increasing and concave upward. This determines the qualitative sketch in part (a).

For rigorous bounds, the alternating Taylor estimates, valid for $x \geq 0$, give

$$1 - x \leq e^{-x} \leq 1 - x + \frac{x^2}{2}.$$

It follows first that $\dot{x} \geq 1$, hence $x(t) \geq t$. The upper inequality gives

$$\dot{x} \leq 1 + \frac{x^2}{2}.$$

Let u solve $u' = 1 + u^2/2$, $u(0) = 0$. Scalar comparison and direct integration yield

$$x(t) \leq u(t) = \sqrt{2} \tan\left(\frac{t}{\sqrt{2}}\right).$$

In particular, on $0 \leq t \leq 1$,

$$x(t) \leq U := \sqrt{2} \tan\left(\frac{1}{\sqrt{2}}\right) \approx 1.208460243.$$

We can sharpen the lower bound using one more Taylor term:

$$e^{-x} \geq 1 - x + \frac{x^2}{2} - \frac{x^3}{6}.$$

Because $0 \leq x(t) \leq U$ for $0 \leq t \leq 1$, we have $x^3 \leq Ux^2$. Therefore

$$\dot{x} \geq 1 + cx^2, \quad c := \frac{1}{2} - \frac{U}{6} > 0.$$

If $\ell' = 1 + c\ell^2$ and $\ell(0) = 0$, comparison gives

$$x(t) \geq \ell(t) = \frac{1}{\sqrt{c}} \tan(\sqrt{c}t).$$

At $t = 1$ this produces the rigorous bracket

$$\boxed{1.113051909 < x(1) < 1.208460243}.$$

Strict inequalities hold for $t > 0$ because the relevant Taylor bounds are strict away from $x = 0$.

For part (c), explicit Euler uses

$$x_{n+1} = x_n + h(x_n + e^{-x_n}), \quad x_0 = 0, \quad qquad h = \frac{1}{N}.$$

A high-accuracy computation gives

$$x(1) \approx 1.1536390027,$$

and therefore, to three decimal places,

$$\boxed{x(1) = 1.154.}$$

Euler's method has global error $O(h)$. Numerically, the error is about $0.2825h$ for small h , so an absolute error below 5×10^{-4} requires h on the order of 10^{-3} . Because the exact value lies fairly close to the rounding

boundary 1.1535, obtaining the displayed rounded value 1.154 reliably requires a somewhat smaller step, approximately $h \lesssim 4 \times 10^{-4}$ (a few times 10^{-4}).

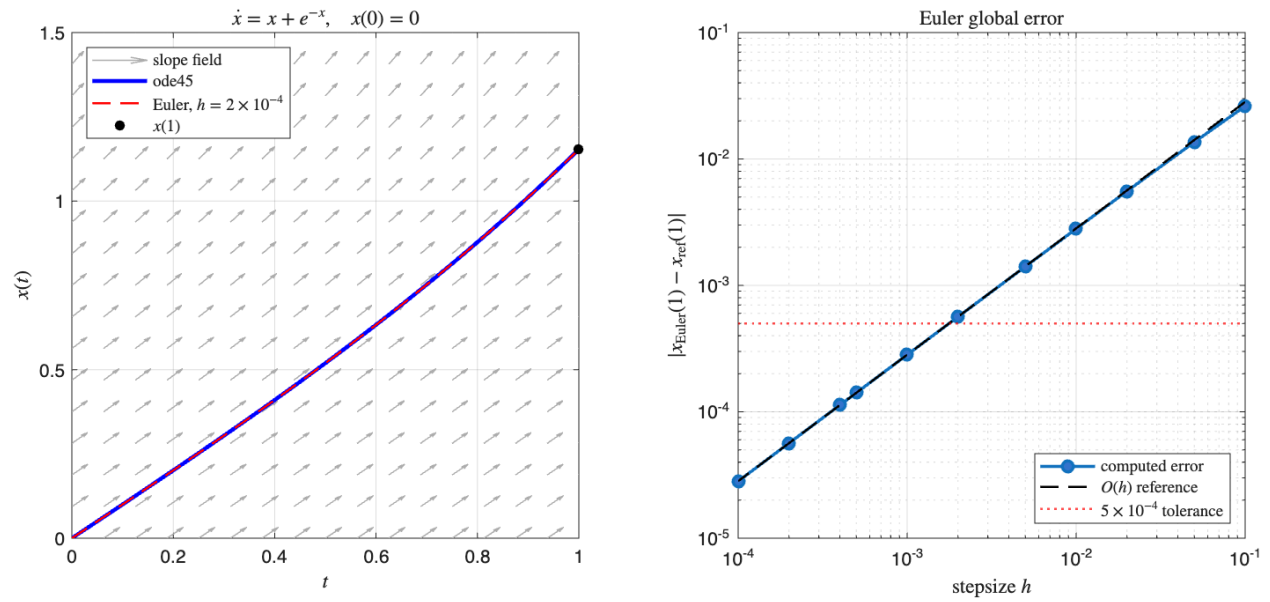


FIGURE 2. MATLAB slope field, numerical solution, and Euler convergence.

MATLAB code. The following script computes the solution, tests Euler stepsizes, plots the slope field and trajectory, and exports the figure used above.

```
% Exercise 2.8.6:  $x' = x + \exp(-x)$ ,  $x(0) = 0$ 
clear; close all; clc;

f = @(t,x) x + exp(-x);

% High-accuracy reference solution.
opts = odeset('RelTol',1e-12,'AbsTol',1e-14);
[tRef,xRef] = ode45(f,[0 1],0,opts);
xExact = xRef(end);

% Explicit Euler approximations at t=1.
Nvals = [10 20 50 100 200 500 1000 2000 2500 5000 10000];
hvals = 1./Nvals;
xEuler = zeros(size(Nvals));
for j = 1:numel(Nvals)
    N = Nvals(j);
    h = 1/N;
    x = 0;
    for n = 1:N
        x = x + h*(x + exp(-x));
    end
    xEuler(j) = x;
end
errors = abs(xEuler-xExact);

fprintf('High-accuracy reference: x(1) = %.12f\n',xExact);
fprintf('      N      h      Euler x(1)      error\n');
fprintf('%8d    %11.4g    %15.10f    %11.4g\n', ...
    [Nvals; hvals; xEuler; errors]);

% A fine Euler trajectory for the solution plot.
Nplot = 5000;
hplot = 1/Nplot;
tE = (0:Nplot)*hplot;
xE = zeros(size(tE));
for n = 1:Nplot
    xE(n+1) = xE(n) + hplot*(xE(n) + exp(-xE(n)));
end
```

```

fig = figure('Color','w','Position',[100 100 1100 450]);

% Solution together with a slope field.
subplot(1,2,1);
tGrid = linspace(0,1,17);
xGrid = linspace(0,1.5,17);
[T,X] = meshgrid(tGrid,xGrid);
S = X + exp(-X);
L = sqrt(1+S.^2);
quiver(T,X,1./L,S./L,0.42,'Color',[0.68 0.68 0.68], ...
        'LineWidth',0.7,'MaxHeadSize',0.25);
hold on;
plot(tRef,xRef,'b-','LineWidth',2.2);
plot(tE,xE,'r--','LineWidth',1.3);
plot(1,xExact,'ko','MarkerFaceColor','k','MarkerSize',5);
xlim([0 1]); ylim([0 1.5]); grid on; box on;
xlabel('$t$', 'Interpreter','latex');
ylabel('$x(t)$', 'Interpreter','latex');
title('$\dot{x}=x+e^{-x}$,\quad $x(0)=0$', 'Interpreter','latex');
legend('slope field','ode45','Euler', '$h=2\times 10^{-4}$','$x(1)$', ...
        'Interpreter','latex','Location','northwest');

% First-order convergence of Euler's method.
subplot(1,2,2);
loglog(hvals,errors,'o-','LineWidth',1.7,'MarkerFaceColor',[0.2 0.45 0.8]);
hold on;
loglog(hvals,errors(end)*(hvals/hvals(end)),'k--','LineWidth',1.2);
yline(5e-4,'r:', 'LineWidth',1.2);
grid on; box on;
xlabel('stepsize $h$', 'Interpreter','latex');
ylabel('$|x_{\rm Euler}(1)-x_{\rm ref}(1)|$', 'Interpreter','latex');
title('Euler global error', 'Interpreter','latex');
legend('computed error','$O(h)$ reference','$5\times 10^{-4}$ tolerance', ...
        'Interpreter','latex','Location','southeast');

if exist('exportgraphics','file') == 2
    exportgraphics(fig,'exercise_2_8_6_matlab.png','Resolution',300);
    exportgraphics(fig,'exercise_2_8_6_matlab.pdf','ContentType','vector');
else

```

```

print (fig , 'exercise_2_8_6_matlab.png', '-dpng', '-r300 ');
print (fig , 'exercise_2_8_6_matlab.pdf', '-dpdf', '-bestfit ');
end

```

```

% ----- OUTPUT -----
%High-accuracy reference:  $x(1) = 1.153639002732$ 
%      N      h      Euler  $x(1)$       error
%      10      0.1      1.1274527973      0.02619
%      20      0.05      1.1400463413      0.01359
%      50      0.02      1.1480752578      0.005564
%      100     0.01      1.1508354119      0.002804
%      200     0.005     1.1522317190      0.001407
%      500     0.002     1.1530747644      0.0005642
%      1000    0.001     1.1533566621      0.0002823
%      2000    0.0005    1.1534977770      0.0001412
%      2500    0.0004    1.1535260133      0.000113
%      5000    0.0002    1.1535824991      5.65e-05
%      10000   0.0001    1.1536107487      2.825e-05

```

□